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ACCESSING THE DESIGN SPACE FOR SOLAR SAILS IN THE EARTH–MOON SYSTEM

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Using a solar sail, a spacecraft orbit can be offset from a central body. Such a trajectory might be desirable for a single-spacecraft relay to support communications with an outpost at the lunar south pole. Although trajectory design within the context of the Earth-Moon restricted problem is advantageous, it is difficult to envision the design space for offset orbits. Numerical techniques to solve boundary-value problems can be employed to understand this new dynamical regime. Numerical finite-difference schemes are typically of low accuracy, but are simple to understand and implement. Two augmented finite-difference methods (AFDMs) are examined in a survey of different sail characteristics and initial guess strategies for the offset orbits. Strategies are presented that exploit knowledge from the AFDM solutions to yield more accurate results using other numerical tools.

INTRODUCTION

In the coming decades, NASA plans to establish a permanent outpost on the Moon for extended human expeditions. Regardless of the specific lunar site that is selected, the astronauts will require a continual communications link between the lunar facility and the Earth. The leading candidate location for the lunar base at this time is at the south pole, which is not always in view of either antennas on the Earth or space-based transceivers, such as the TDRSS satellites, in geosynchronous orbit. Therefore, two or more spacecraft in traditional polar orbits about the Moon must alternate responsibility as relays for the Earth–Moon link.[1]

In contrast to a constellation of multiple spacecraft, alternative communication strategies that rely on only one satellite do exist, using current or near-future technology. Advanced propulsion concepts, such as low-thrust ion engines as well as solar sails, supply a force in addition to gravity and can actually offset an orbit from the Moon.[2,3,4] Such an orbit allows one spacecraft to be in view of a point on the lunar surface at all times. This dynamical system is complex due to the gravitational effects of the two primaries and, in the case of solar sails, the fact that the Sun, which influences the direction and magnitude of the resulting sail thrust vector, continually shifts position relative to the Earth–Moon system. A previous approach, based on an understanding of the dynamical structure and using that knowledge to design an orbit, has been successful in developing solar sail orbits in the vicinity of the lunar Lagrange points.[5,6,7,8,9] However, motion about one of the lunar poles requires an alternative strategy.

When developing spacecraft trajectories to support various mission scenarios, designers frequently use intuition as the basis for the orbit concept. Keplerian ellipses are appropriate approximations in two-body systems with small perturbations. Dynamical systems theory can be employed to supply a dynamical framework in a multi-body regime where transfers between orbits in the vicinity of Lagrange points are designed within the context of the circular restricted three-body (CR3B) system.[10,11] Both of these approaches are easily transitioned to full ephemeris models, and the resulting trajectories resemble orbits from the lower-fidelity models.

Ion propulsion and solar sails supply an additional force to the spacecraft, often expanding the design options. However, envisioning the necessary control history and the resulting trajectory prior to any analysis can be difficult, especially when the advanced-propulsion device is employed in a more complex dynamical

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regime, such as a CR3B system. Analytical techniques that incorporate these control forces in such a regime are not sufficiently developed to quickly reveal the extent of the design possibilities.

Numerical methods are critical elements in uncovering new trajectory options and providing mission designers a better sense of the design space, particularly in complex regimes. Spacecraft trajectories are solutions to nonlinear equations of motion (EOMs), which themselves are ordinary differential equations (ODEs). Some ODEs are initial value problems (IVPs) and can be solved numerically using explicit or implicit integration, incorporating single or multiple steps. In an IVP, the state is defined at some point in the domain (time) of the solution to the differential equations, and all subsequent states along the trajectory arise from the differential equations and the given initial condition. For explicit integration, the state at the current time is calculated based on the state at the previous time (or, in a multi-step method, at a number of preceding times); it is the method of integration most analysts use regularly. Implicit integration may also be single or multi-step and also relies on the initial conditions and the differential equations, but the state at the current time is expressed implicitly (i.e., within) the generating equation. An iterative method, such as Newton-Raphson, is required to solve the differential equations at the current point in the time domain. Implicit methods are generally more difficult to solve, but the accuracy of the solution is higher for stiff ODEs.[12,13]

Trajectories can also be represented as solutions to boundary value problems (BVPs) in terms of ordinary differential equations (ODEs), in which conditions are specified at the beginning (like the IVP) and, also, at the end of the time domain. For periodicity, the conditions, or states, at the beginning of the trajectory must equal those at the end. The specific values of the states at the extremes may or may not be fixed in a BVP with a periodicity constraint; periodicity only requires that the values at the extremes are equal. Differential correctors[14] and shooting methods (adaptations of the IVP), collocation, and finite-difference methods[15,12,16] are common numerical tools to solve BVPs. Variational methods are also used to solve BVPs when the problem possesses certain minimality properties.[13]

Both a shooting method and a differential corrector can be formulated simply as an initial value problem with a constrained initial position and an initial velocity that is free. The difference between the end state and the target state at the terminal point on the trajectory arc is used to adjust the initial velocity.[12,14,11,17] Shooting is often used in the optimization of spacecraft trajectories.[18,19] Unfortunately, continuous control, such as a solar sail, cannot be easily incorporated into shooting methods and differential correctors. Trajectory arcs involving low-thrust ion engines are typically discretized such that the trust arcs resemble a series of finite burns.

Collocation schemes have arisen in the past few decades as an alternative strategy to generate spacecraft trajectories, coincident with improvements in computer processor capabilities.[20,21,22,23,2] The advantage of a collocation scheme in solving a BVP is a formulation such that the control can be discretized; then the algorithm delivers a solution for the control and trajectory states simultaneously.[24] Collocation involves minimization of the difference between the derivative of a continuous polynomial approximation and the derivative from the equations of motion at an intermediate, or *defect*, point between nodes along a trajectory. Mathematically, collocation methods require selection of a representation (typically a polynomial) that is consistent with the vector field of the differential equations.[25] Note that implicit Runge-Kutta techniques are equivalent to collocation schemes;[25,26] however, the present discussion centers on collocation as a means to solve BVPs. In the 1970's, collocation methods were applied to boundary value problems by Russell and Shampine.[24] Doedel also approached solving nonlinear two-point boundary problems using finite differences in a collocation strategy.[27]

In the mid-1980's, Hargraves and Paris[20] applied a collocation method to a direct trajectory optimization problem. A functional (indirect) optimization problem is transcribed to a parameter (direct) optimization problem by using discretized interior states as optimization parameters. Nonlinear programming techniques are then employed to solve for the set of interior states that optimizes some cost function, usually fuel or final time.[20,21,22] Herman and Conway[23] improved the accuracy of the technique by incorporating high-degree Gauss-Lobatto quadrature rules into a collocation and nonlinear-programming algorithm. Tang and Conway[28] applied these techniques to low-thrust interplanetary trajectories, and Melton[29] successfully adapted them to the interplanetary solar sail problem. Other authors have also explored trajectory optimization as a parameter

optimization problem.[30,19] The focus of the preceding studies is on direct methods, which are generally less accurate than indirect methods,[18] but could be used as starting points for indirect optimization schemes.

Most recently, Ozimek, Grebow, and Howell applied low- and high-degree Gauss-Lobatto rules to generate solar sail trajectories in the Earth–Moon system.[2] The local error along the trajectory path was proportional to Δt^5 and Δt^{12}, respectively, due to truncation in the approximating polynomials.[31] However, the problem was not formulated as an optimization problem. Rather, the authors employed this approach to iteratively solve the equations of motion. The converged solution rarely resembles the initial guess; however, the solution does depend on the initial guess.

The preceding numerical methods possess high degrees of accuracy. However, shooting methods and differential correctors require some knowledge of the design space to initiate the procedure. Collocation schemes do not require a priori knowledge of the solution corresponding to the BVP, but the implementation is not intuitive. Furthermore, the partial derivatives of the constraints with respect to the states are nontrivial in collocation schemes and are often approximated numerically.[2] A more rudimentary method for solving BVPs is the finite-difference method (FDM), in which the derivatives that appear in the differential equations are replaced with their respective finite differences and evaluated at node points along the trajectory.[12] The solution process is clearly iterative. The trajectory is discretized and the equations that represent the relationships at the nodes are solved simultaneously. Unlike the collocation method, where an approximation is employed to solve the equations of motion, the FDM approximates the equations of motion, then produces a solution to that approximation. Using a central difference approximation, the best accuracy available from a FDM is proportional to Δt^2. However, the FDM is simple to formulate and offers an improved understanding of the design space. Once the analyst is familiar with the feasible options, more accurate methods for solving the BVP can be combined with optimization techniques to produce best-estimate trajectories.

In the following sections, a FDM is described and applied to solar sails in the Earth–Moon CR3B system. Also included is a modified FDM for the same application, where the velocity is included as part of the solution at each node along the discretized trajectory. This second scheme was used to create the orbits in Wawrzyniak and Howell.[3] Neither method is strictly a straightforward textbook FDM,[15,12,16] since both are augmented with trajectory and control constraints. Additionally, in contrast to a simple two-point BVP, where the states are fixed at the extremes, the trajectory is required to be periodic; nowhere along the arc is the solution pre-specified.

A description of the dynamical model in the Earth–Moon CR3B system is followed by an algorithm using the augmented FDMs for generating trajectories. The strategies are based on minimizing the difference between accelerations from the equations of motion (and velocities in the case of the second augmented FDM) and the corresponding values numerically derived from positions along a discretized trajectory. Different options to generate initial guesses for the trajectory (two) and for the control (six) lead to different results. Given the low accuracy of these methods, an error analysis is included. Finally, a survey of solutions illustrates the expanded design space and the required sail characteristics for continuous coverage of the lunar south pole using a solar sail spacecraft.

SYSTEM DYNAMICAL MODEL

The problem that represents this application is defined within the context of the CR3B system, which is modeled in a frame, b, that is rotating with respect to an inertial system, e. A CR3B model that incorporates the gravity contributions of two primary bodies is also geometrically advantageous for understanding the problem. Following McInnes,[32] the nondimensional vector equation of motion for a spacecraft at a location $\mathbf{r}$ relative to the barycenter (center of mass of the primaries) is

$$\mathbf{a} + 2\left(\boldsymbol{\omega} \times \mathbf{v}\right) + \boldsymbol{\nabla} U(\mathbf{r}) = \mathbf{a}_s(t) \tag{1}$$

where the first term is the acceleration relative to the rotating frame (more precisely expressed as $\frac{{}^b d^2\mathbf{r}}{dt^2}$) and the second term is the corresponding Coriolis acceleration, which requires the velocity relative to the rotating frame, $\mathbf{v}$ (more precisely $\frac{{}^b d\mathbf{r}}{dt}$).* The angular-velocity vector, $\boldsymbol{\omega}$ (or ${}^e\boldsymbol{\omega}^b$), relates the orientation of the rotating

*Vectors are denoted with boldface.

frame to the inertial frame. The applied acceleration, from a solar sail in this case, is represented on the right side by $\mathbf{a}_s(t)$. The pseudo-gravity gradient, $\boldsymbol{\nabla} U(\mathbf{r})$, combines centripetal and gravitational acceleration

$$\boldsymbol{\nabla} U(\mathbf{r}) = \Big(\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \Big) + \left(\frac{(1-\mu)}{r_1^3} \mathbf{r_1} + \frac{\mu}{r_2^3} \mathbf{r_2} \right) \tag{2}$$

where μ represents the mass fraction of the smaller body and r_1 and r_2 are the distances from the larger and smaller bodies, respectively, that is,

$$r_1 = \sqrt{(\mu + x)^2 + y^2 + z^2}$$
$$r_2 = \sqrt{(\mu + x - 1)^2 + y^2 + z^2}$$

At a distance of 1 AU, an appropriate distance to assume for a sailcraft in the Earth–Moon system, the applied acceleration from a solar sail is modeled as

$$\mathbf{a}_s(t) = \beta (\hat{\boldsymbol{\ell}}(t) \cdot \mathbf{u})^2 \mathbf{u} \tag{3}$$

where $\mathbf{u}$ is the sailface normal, $\hat{\boldsymbol{\ell}}(t)$ is a unit vector in the sunlight direction, and β is the sail's characteristic acceleration in nondimensional units. The term $(\hat{\boldsymbol{\ell}}(t) \cdot \mathbf{u})$ is also expressed as $\cos^2 \alpha$, where α is the sail pitch angle, or the angle between the solar incidence direction and the sail normal. To generate the sail acceleration in dimensional units, a_0, β is multiplied by the system characteristic acceleration, a^*, that is, the relationship between the dimensional and nondimensional acceleration in Eq. (1). In fact a^* is the ratio of the characteristic length, L^* (384,400 km for the Earth–Moon distance), to the square of the characteristic time, t^* ($2\pi t^* = 27.321$ days), that is,[3]

$$a^* = \frac{L^*}{t^{*2}} = 2.7307 \text{ mm/s}^2$$

The current limits of solar sail technology place the sail characteristic acceleration, a_0, between 0.58 and 1.70 mm/s^2 (0.212 to 0.623 nondimensional acceleration units), assuming the sail and structural components can be scaled while retaining the same payload and control system.[33]

The sunlight direction is expressed relative to the rotating frame and is a function of time, that is,

$$\hat{\boldsymbol{\ell}}(t) = \cos(\Omega t)\hat{\mathbf{b}}_1 - \sin(\Omega t)\hat{\mathbf{b}}_2 + 0\hat{\mathbf{b}}_3 \tag{4}$$

where Ω is the ratio of the synodic rate of the Sun as it moves along its path to the system rate, approximately 0.9192. One physical constraint is imposed on the attitude of the spacecraft: the sail normal, $\mathbf{u}$, which is coincident with the direction of the resultant force, is always directed away from the Sun. This is written mathematically as

$$\hat{\boldsymbol{\ell}}(t) \cdot \mathbf{u} \geq 0 \tag{5}$$

This dot product can also be expressed as $(\cos \alpha)$, where α is the sail pitch angle.

THE AUGMENTED FINITE-DIFFERENCE METHODS

A generic, nonlinear, second-order, two-point BVP can be represented as

$$\ddot{\mathbf{r}} = f(t, \mathbf{r}, \dot{\mathbf{r}}) \tag{6}$$

where $\mathbf{r}(t_1)$ and $\mathbf{r}(t_n)$ are fixed at the extremes of the time span, $[t_1, t_n]$.[15,12,16,13] The solution of Eq. (6) is a trajectory, $\mathbf{r}(t)$. To solve the BVP, the classic FDM discretizes the domain into nodes, or epochs, $t_1, t_2, \ldots, t_n$ and replaces the derivatives in Eq. (6) by their respective finite differences. Central-difference approximations (CDAs) are commonly used because they are more accurate than forward or backward differences. The CDAs for the first and second time derivatives of $\mathbf{r}(t)$ are, of course, velocity and acceleration, and are approximated by CDAs as

$$\mathbf{v}(t) \approx \frac{\mathbf{r}_{i+1} - \mathbf{r}_{i-1}}{t_{i+1} - t_{i-1}} \tag{7}$$

$$\mathbf{a}(t) \approx 2\frac{(\mathbf{r}_{i+1}-\mathbf{r}_i)(t_i-t_{i-1})-(\mathbf{r}_i-\mathbf{r}_{i-1})(t_{i+1}-t_i)}{(t_i-t_{i-1})(t_{i+1}-t_{i-1})(t_{i+1}-t_i)} \tag{8}$$

where $\mathbf{r}(t_i)$ is defined as $\mathbf{r}_i$ and $t_i \in (t_{i-1}, t_{i+1})$. Equation (8) arises from three central differences: one for the velocity at $t_{i-1/2}$, which is midway between t_{i-1} and t_i, another at $t_{i+1/2}$, midway between t_i and t_{i+1} and a third using the first two intermediate velocities resulting in the acceleration at t_i. The FDM does not require uniform node placement*, but when the time steps between t_{i-1}, t_i, and t_{i+1} are fixed at a value of Δt, then the midpoint of $[t_{i-1}, t_{i+1}]$ is t_i and Eqs. (7) and (8) simplify to

$$\widetilde{\mathbf{v}}_i = \frac{\mathbf{r}_{i+1}-\mathbf{r}_{i-1}}{2\Delta t} \tag{9}$$

$$\widetilde{\mathbf{a}}_i = \frac{\mathbf{r}_{i+1}-2\mathbf{r}_i+\mathbf{r}_{i-1}}{\Delta t^2} \tag{10}$$

where the symbol "$\widetilde{\cdot}$" indicates a numerical approximation.† The differences between the actual velocity and acceleration and their respective CDAs at t_i are

$$\mathbf{v}_i - \widetilde{\mathbf{v}}_i = -\frac{\Delta t^2}{6}\mathbf{r}'''(\rho_v) \tag{11}$$

$$\mathbf{a}_i - \widetilde{\mathbf{a}}_i = -\frac{\Delta t^2}{12}\mathbf{r}^{\mathrm{iv}}(\rho_a) \tag{12}$$

where $\rho_{v,a} \in (t_{i-1}, t_{i+1})$ and $\mathbf{r}'''(t)$ and $\mathbf{r}^{\mathrm{iv}}(t)$ are continuous on $[t_{i-1}, t_{i+1}]$. Derivations of these truncation errors are presented in the literature.[15,12,16] The approximations in Eqs. (9) and (10) replace the first and second time derivatives in Eq. (6), and the resulting form is an equation at each epoch of the form

$$\frac{\mathbf{r}_{i-1}+2\mathbf{r}_i+\mathbf{r}_{i+1}}{\Delta t^2} = f\left(t, \mathbf{r}, \frac{\mathbf{r}_{i+1}-\mathbf{r}_{i-1}}{2\Delta t}\right) \tag{13}$$

Equation (13) may still be nonlinear, but it can be linearized into a system of equations, one for each t_i, excluding the extremes, and solved iteratively. Due the approximations in Eqs. (9) and (10), the classic FDM results in a solution that will approximate each component of the true solution to the BVP by an error proportional to Δt^2 at each node.[15,16]

For the current problem, the FDM was augmented to incorporate path and system constraints, as well as a control history. This method is labeled AFDM. Additionally, although BVPs usually require specific values for some or all of their boundary conditions, the AFDM requires only that the first and last state be equal for periodicity. Two variations of the AFDM are also developed. The first is similar to the FDM, where position is the only explicitly discretized trajectory state (AFDM-R). The second AFDM is formulated to directly solve for position and velocity along the discretized trajectory (AFDM-RV). Algebraically, these AFDMs are equivalent. In practice, the two formulations yield slightly different results. Given the goal to present a simple and quick method to approximate the design space, neither should be considered superior to the other, but both achieve the stated goal.

The AFDM-R is formulated by differencing the analytical acceleration from Eq. (1) and the CDA of acceleration from Eq. (9) and (10), that is,

$$\widetilde{\mathbf{a}}_i + 2(\boldsymbol{\omega} \times \widetilde{\mathbf{v}}_i) + \boldsymbol{\nabla} U(\mathbf{r}_i) - \mathbf{a}_s(t_i) = 0 \tag{14}$$

Note that in Eq. (14) the velocity term is replaced with its CDA, $\widetilde{\mathbf{v}}_i$. To solve the BVP, the trajectory is discretized at $t_1, t_2 \ldots, t_n$. Position is expressed in terms of Cartesian coordinates at each node,

$$\mathbf{r}_i = \begin{Bmatrix} x_i & y_i & z_i \end{Bmatrix}^{\mathrm{T}} \tag{15}$$

*Mesh refinement and non-uniform node spacing for improved accuracy of the FDM is a potential area of further investigation.

†It should be noted that $\mathbf{r}_{n-1}$ is used for $\mathbf{r}_0$ when calculating $\widetilde{\mathbf{v}}_1$ for a periodic solution. Similarly, $\mathbf{r}_1$ substitutes for $\mathbf{r}_n$ in $\widetilde{\mathbf{v}}_{n-1}$, so the problem is not over-constrained. The calculations for $\mathbf{a}_1$ and $\mathbf{a}_{n-1}$ are similar.

The control, $\mathbf{u}_i$, which is the pointing vector of the solar sail in this application, and a set of m slack variables, η_i, are also included at each node. The positions, control, and slack variables at node i for the AFDM-R are collected into the subvector,

$$\mathbf{q}_i = \left\{ \begin{array}{cccc} \mathbf{r}_i^{\mathrm{T}} & \mathbf{u}_i^{\mathrm{T}} & \boldsymbol{\eta}_i^{\mathrm{T}} \end{array} \right\}_{(6+m)}^{\mathrm{T}} \tag{16}$$

where the length of $\mathbf{q}_i$ is $(6+m)$. The subvectors associated with each node are collected into a column vector,

$$\mathbf{X} = \left\{ \begin{array}{ccccc} \mathbf{q}_1^{\mathrm{T}} & \mathbf{q}_2^{\mathrm{T}} & \cdots & \mathbf{q}_{n-1}^{\mathrm{T}} & \mathbf{q}_n^{\mathrm{T}} \end{array} \right\}_{(6+m)n}^{\mathrm{T}} \tag{17}$$

which represents the discretized trajectory as well as the associated control history and slack variables.

The AFDM-RV is similarly formulated, except that there are two distinct sets of ODEs to solve, i.e.,

$$\widetilde{\mathbf{v}}_i - \mathbf{v}_i = 0 \tag{18}$$

$$\widetilde{\mathbf{a}}_i + 2\left(\boldsymbol{\omega} \times \mathbf{v}_i\right) + \boldsymbol{\nabla} U(\mathbf{r}_i) - \mathbf{a}_s(t_i) = 0 \tag{19}$$

Note the different velocity terms in Eqs. (14) and (19). With the AFDM-RV, the subvector is

$$\mathbf{q}_i = \left\{ \begin{array}{cccc} \mathbf{r}_i^{\mathrm{T}} & \mathbf{v}_i^{\mathrm{T}} & \mathbf{u}_i^{\mathrm{T}} & \boldsymbol{\eta}_i^{\mathrm{T}} \end{array} \right\}_{(9+m)}^{\mathrm{T}} \tag{20}$$

since $\mathbf{v}_i$ is now explicitly part of the solution. This formulation results in an $\mathbf{X}$ that is $(9+m)n$.

Both the AFDM-R and the AFDM-RV algorithms constrain the difference between the evaluated EOMs at node i and the associated numerically derived approximations to be zero. Additional constraints are required for periodicity, control-direction magnitude, and path characteristics. All of these constraints are dependent on $\mathbf{X}$ and are collected in the column vector $\mathbf{F}(\mathbf{X})$. An expansion of $\mathbf{F}(\mathbf{X})$ about $\mathbf{X}^{j+1}$ yields a linear approximation for $\mathbf{F}(\mathbf{X}^{j+1})$, that is,

$$\mathbf{F}(\mathbf{X}^{j+1}) = \mathbf{F}(\mathbf{X}^{j}) + D\mathbf{F}(\mathbf{X}^{j})\left(\mathbf{X}^{j+1} - \mathbf{X}^{j}\right) + \mathscr{O}(\Delta \mathbf{X}^2) \tag{21}$$

where the $D\mathbf{F}(\mathbf{X}^j)$ is the Jacobian, $\frac{\partial \mathbf{F}(\mathbf{X}^j)}{\partial \mathbf{X}^j}$.*† By assuming that $\mathbf{X}^j$ is in the neighborhood of $\mathbf{F}(\mathbf{X}) = 0$ (i.e., $\|\mathbf{X}^{j+1} - \mathbf{X}^j\| \ll 1$), Eq. 21 is rearranged into a least-norm problem, that is,

$$\mathbf{X}^{j+1} = \mathbf{X}^{j} - D\mathbf{F}(\mathbf{X}^{j})^{\mathrm{T}}\left[D\mathbf{F}(\mathbf{X}^{j}) \cdot D\mathbf{F}(\mathbf{X}^{j})^{\mathrm{T}}\right]^{-1}\mathbf{F}(\mathbf{X}^{j}) \tag{22}$$

Equations (21) and (22) reflect the Newton-Raphson method of root finding for several variables. For efficient computation, $D\mathbf{F}(\mathbf{X})$ is pre-allocated and stored in memory as a sparse matrix.[34] Equation (22) is iterated until some convergence tolerance is met,

$$\frac{\|\mathbf{X}^{j+1} - \mathbf{X}^{j}\|}{\|\mathbf{X}^{j}\|} \leq \mathit{tol}$$

When an initial guess is in the neighborhood of the solution, convergence is quadratic. Specifying $\mathit{tol} \approx 1 \times 10^{-7}$ is sufficient, since any further iteration approaches double precision. If the initial guess is not in the neighborhood, $\mathbf{X}^{j+1}$ will possess little resemblance to $\mathbf{X}^j$; the step from $\mathbf{X}^j$ to $\mathbf{X}^{j+1}$ may even be chaotic. However, the new $\mathbf{X}^{j+1}$ might be in the neighborhood of an alternate solution and subsequently lead to convergence. The converged discretized trajectory, $\mathbf{X}^f$, that satisfies these constraints is the trajectory that solves the EOMs.

At each epoch, the converged subvector $\mathbf{q}_i$ will differ from the true solution at that epoch, $\overline{\mathbf{q}}_i$, due to the truncation error of the AFDMs and limited machine precision. The proceeding error analysis is described by Kincaid and Cheney [16, pp. 589–592]. For a Δt greater than some value, the error due to truncation in

*Superscripts on vectors refer to iteration number. The initial guess is denoted $j = 0$; a converged solution by $j = f$.

†Partial derivatives for the Jacobian are not derived here.

Eqs. (7)–(12) will dominate the machine error. Assuming that there exists a true set of position states, $\bar{\mathbf{r}}$, that solve Eqs. (1) and (11) for the AFDM-R, Eq. (6) can be written as

$$\frac{\bar{\mathbf{r}}_{i-1} + 2\bar{\mathbf{r}}_i + \bar{\mathbf{r}}_{i+1}}{\Delta t^2} - \frac{\Delta t^2}{12}\bar{\mathbf{r}}^{\text{iv}}(\rho_a) = -\frac{\boldsymbol{\omega} \times (\bar{\mathbf{r}}_{i+1} - \bar{\mathbf{r}}_{i-1})}{\Delta t} + \frac{\Delta t^2}{6}\boldsymbol{\omega} \times \bar{\mathbf{r}}'''(\rho_v) - \boldsymbol{\nabla} U(\bar{\mathbf{r}}_i) + \mathbf{a}_s(t_i) \quad (23)$$

The pseudo-gravity gradient associated with the true solution, $\bar{\mathbf{r}}_i$, can be expanded as a function of an approximate solution, $\mathbf{r}_i$, that is,

$$\boldsymbol{\nabla} U(\bar{\mathbf{r}}_i) = \boldsymbol{\nabla} U(\mathbf{r}_i) + \mathbf{M}(\bar{\mathbf{r}}_i)\mathbf{e}_i + \mathscr{O}(\mathbf{e}_i^{\mathrm{T}}\mathbf{e}_i) \quad (24)$$

where $\mathbf{e}_i = \bar{\mathbf{r}}_i - \mathbf{r}_i$ and $\mathbf{M}(\bar{\mathbf{r}}_i)$ is the Hessian of U. Using the solution from the AFDM-R, Eq. (13) is subtracted from Eq. (23), resulting in

$$\Delta t^{-2}\left(\mathbf{e}_{i-1} + 2\mathbf{e}_i + \mathbf{e}_{i+1}\right) = -\Delta t^{-1}\left(\boldsymbol{\omega} \times (\mathbf{e}_{i+1} - \mathbf{e}_{i-1})\right) - \mathbf{M}_i\mathbf{e}_i + \Delta t^2\mathbf{h}_i \quad (25)$$

where $\mathbf{h}_i = \frac{1}{12}\bar{\mathbf{r}}^{\text{iv}}(\rho_a) + \frac{1}{6}\bar{\mathbf{r}}'''(\rho_v)$. It is assumed that $\bar{\mathbf{r}}(t)$ is continuously differentiable to fourth order. Rearranging and multiplying by Δt^2 results in

$$(-\mathbf{I} + \Delta t\boldsymbol{\omega}\times)\mathbf{e}_{i-1} + (2\mathbf{I} - \Delta t^2\mathbf{M}_i)\mathbf{e}_i + (-\mathbf{I} - \Delta t\boldsymbol{\omega}\times)\mathbf{e}_{i+1} = -\Delta t^4\mathbf{h}_i \quad (26)$$

where $\mathbf{I}$ is the identity matrix and $\boldsymbol{\omega}\times$ is the skew-symmetric gyroscopic matrix. Some terms in Eq. (26) can be simplified to a form that will be useful later, that is,

$$(-\mathbf{I} + \Delta t\boldsymbol{\omega}\times) + (2\mathbf{I} - \Delta t^2\mathbf{M}_i) + (-\mathbf{I} - \Delta t\boldsymbol{\omega}\times) = -\Delta t^2\mathbf{M}_i \quad (27)$$

Now, let $\boldsymbol{\lambda}$ correspond to the $\mathbf{e}_i$ with the largest magnitude. Equation (26) reduces to an upper bound on $\boldsymbol{\lambda}$,

$$(-\mathbf{I} + \Delta t\boldsymbol{\omega}\times)\boldsymbol{\lambda} + (2\mathbf{I} - \Delta t^2\mathbf{M}_i)\boldsymbol{\lambda} + (-\mathbf{I} - \Delta t\boldsymbol{\omega}\times)\boldsymbol{\lambda} \geq -\Delta t^4\mathbf{h}_i \quad (28)$$

Using Eq. (27), Eq. (28) simplifies to

$$\begin{aligned} -\Delta t^2\mathbf{M}_i\boldsymbol{\lambda} &\geq -\Delta t^4\mathbf{h}_i \\ \mathbf{M}_i\boldsymbol{\lambda} &\leq \Delta t^2\mathbf{h}_i \\ \boldsymbol{\lambda} &\leq \Delta t^2\mathbf{M}_i^{-1}\mathbf{h}_i \end{aligned} \quad (29)$$

where the inequality is with respect to the corresponding components of the vectors on each side; the inequalities in Eq. (29) represent three scalar equations. Thus, the position error for the AFDM-R is $\mathscr{O}(\Delta t^2)$ as $\Delta t \to 0$. To calculate the velocity from the AFDM-R, the CDA from Eq. (9) is used, also with an error $\mathscr{O}(\Delta t^2)$. The error analysis for the AFDM-RV algorithm incorporates the fact that $\mathbf{v} = f(t, \mathbf{r})$ has errors that can also be expressed by Eq. (11), and, therefore, the derivation continues with the same steps as those for the AFDM-R beginning with Eq. (23).

The AFDMs sacrifice accuracy for simplicity. At each node, the error in the trajectory is proportional to Δt^2, significantly less accurate than common collocation methods.[20,31] To improve the accuracy of the AFDMs, smaller step sizes can be used. In this analysis, $\Delta t = 0.067$, or $n = 101$, which translates to an error on the order of 4.52×10^{-3} Earth–Moon distances, approximately 1740 km, in each direction.

To achieve the local Δt^5 accuracy of a Hermite-Simpson collocation method, Δt must equal 0.0012, or $n = 5800$. The Jacobian $D\mathbf{F}(\mathbf{X})$ becomes quite large, imposing a significant computational cost. Furthermore, the smaller the step size, the larger the machine error.[12] Incorporating a process to determine the step size that minimizes some combination of truncation and machine errors, adds complexity to a method that is formulated to be simple to understand and implement, as well as quick to yield results. Nonetheless, prior to any analysis, the design space for this problem is unknown, so a trajectory that is within Δt^2 of an actual solution is meaningful. Since initial conditions are not specified and only periodicity is required, it is possible that the Δt^2 trajectory used as an initial guess in a more accurate methods would converge on a neighboring trajectory. If the goal is a general understanding of the design space accomplished in a relatively quick analysis, then the results can be very insightful.

ALGEBRAIC CONSTRAINT VECTOR, F(X)

At the core of the AFDM algorithms is the algebraic constraint vector, $\mathbf{F}(\mathbf{X})$, which contains the algebraic approximations to the equations of motion and other constraints necessary for the desired periodic solar sail trajectory. Each element in $\mathbf{F}(\mathbf{X})$ should be zero to satisfy the ODEs and path constraints. For a periodic solar sail trajectory, subject to path constraints $\mathbf{g}(\mathbf{X})$,

$$\mathbf{F}(\mathbf{X}) = \left\{ \begin{array}{ccccccc} \Delta\mathbf{a}(\mathbf{X})^{\mathrm{T}} & [\Delta\mathbf{v}(\mathbf{X})^{\mathrm{T}}] & \mathbf{T}(\mathbf{X})^{\mathrm{T}} & y_1 & \mathbf{N}(\mathbf{X})^{\mathrm{T}} & \mathbf{g}(\mathbf{X})^{\mathrm{T}} \end{array} \right\}^{\mathrm{T}} \tag{30}$$

Again, the presence of $\Delta\mathbf{v}(\mathbf{X})$ depends on whether the method is the AFDM-R or AFDM-RV, and the length of $\mathbf{F}(\mathbf{X})$ will be either $(2 + (4 + m)n)$ or $(2 + (7 + m)n)$, accordingly. The order of these elements within the vector is arbitrary and no difference in performance is apparent when $\mathbf{F}(\mathbf{X})$ and $\mathbf{X}$ are rearranged into a banded-diagonal matrix. Using the configuration in Eq. (30), each element set appears as block diagonal in the corresponding Jacobian, $D\mathbf{F}(\mathbf{X})$. Each set is subsequently discussed. The corresponding size of each set is indicated with a subscript outside of the braces.

Acceleration at a given epoch, i, as evaluated from the equations of motion depends only on the position, velocity, and control at that epoch, $\mathbf{a}_i = \mathbf{f}(\mathbf{r}_i, \mathbf{v}_i, \mathbf{u}_i)$. The numerically derived acceleration, $\widetilde{\mathbf{a}}_i$ from Eq. (8) or (12), depends on the state at multiple epochs. A valid trajectory possesses the same acceleration whether computed from the EOMs or from the CDAs, within the truncation error. Therefore, the difference between accelerations resulting from the EOMs and those from numerical calculation should be zero; thus, the difference forms the first set of equality constraints in the composite constraint vector,

$$\Delta\mathbf{a}(\mathbf{X}) = \left\{ \begin{array}{c} \mathbf{a}_1(\mathbf{r}_1, \mathbf{v}_1, \mathbf{u}_1) - \widetilde{\mathbf{a}}_1 \\ \vdots \\ \mathbf{a}_{n-1}\left((\mathbf{r}, \mathbf{v}, \mathbf{u})|_{n-1}\right) - \widetilde{\mathbf{a}}_{n-1} \end{array} \right\}_{3(n-1)} \tag{31}$$

Likewise, the difference between the velocity from the EOMs and the numerically derived velocity is zero for a valid trajectory. Velocity from the EOMs at each epoch, $\mathbf{v}_i$, is available from $\mathbf{X}$ (see Eqs. (17) and (20)). This difference forms a second set of equality constraints (only in the AFDM-RV algorithm) in $\mathbf{F}(\mathbf{X})$,

$$\Delta\mathbf{v}(\mathbf{X}) = \left\{ \begin{array}{c} \mathbf{v}_1 - \widetilde{\mathbf{v}}_1 \\ \vdots \\ \mathbf{v}_{n-1} - \widetilde{\mathbf{v}}_{n-1} \end{array} \right\}_{3(n-1)} \tag{32}$$

These differences are also denoted as *defects* and are located at the node at each t_i. In an initial guess for the trajectory state vector, the defects $\Delta\mathbf{a}_i$ and $\Delta\mathbf{v}_i$ at each node will most likely not be zero. The Newton-Raphson iteration process adjusts the path of the trajectory to resolve these differences. Figure 1 illustrates an example of the acceleration and velocity defects before and after the corrections process. The arrows indicate the first $(\mathbf{v}, \widetilde{\mathbf{v}})$ and second $(\mathbf{a}, \widetilde{\mathbf{a}})$ derivatives along the trajectory path by evaluating the EOMs $(\mathbf{v}, \mathbf{a})$ and the numerical approximation $(\widetilde{\mathbf{v}}, \widetilde{\mathbf{a}})$ from the candidate trajectory (the top curve). The differences between the corresponding arrows represent the defects $(\Delta\mathbf{v}_i, \Delta\mathbf{a}_i)$. After the correction, it is apparent that the CDAs of the position and velocity closely match the respective EOMs. (The process in the figure is the AFDM-RV, where velocity is directly solved in addition to position.)

To enforce periodicity, the goal is a first and last state vector that are equal, which is represented as a set of constraints in $\mathbf{F}(\mathbf{X})$ as

$$\mathbf{T}(\mathbf{X}) = \left\{ \begin{array}{c} \mathbf{r}_n - \mathbf{r}_1 \\ [\mathbf{v}_n - \mathbf{v}_1] \\ \mathbf{u}_n - \mathbf{u}_1 \\ \boldsymbol{\eta}_n - \boldsymbol{\eta}_1 \end{array} \right\}_{(6|9+m)} \tag{33}$$

where m is the number of path constraints at the first and last node, expressed as slack variables. The length of $\mathbf{T}(\mathbf{X})$ depends on the whether the method is AFDM-R or AFDM-RV.

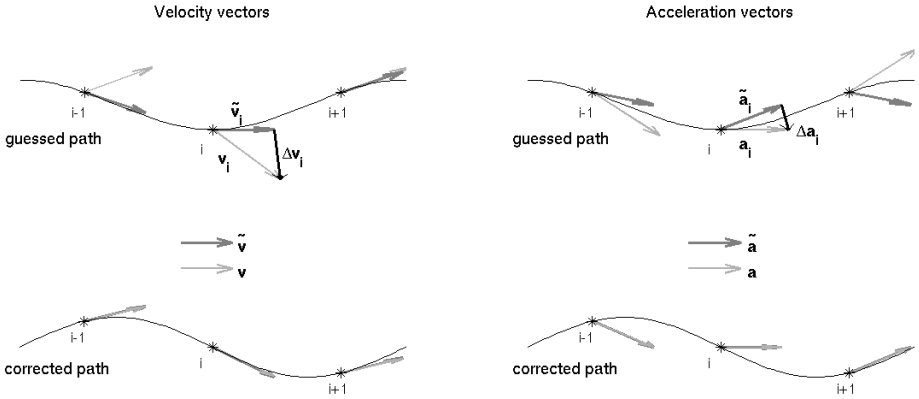


Figure 1. An example trajectory before (top) and after (bottom) the numerical corrections process.

Although the only boundary condition is periodicity, for consistency the position at the boundary is constrained to be in the xz-plane. Therefore, the y-coordinate at the first point along the trajectory is constrained, $y_1 = 0$. In practice, without this constraint, y_1 remained within approximately 1×10^{-7} nondimensional distance units at the converged solution. With this constraint, y_1 is reduced to $\pm 1 \times 10^{-14}$ nondimensional distance units.

Next, the magnitude of the control vector must remain of unit length in the iteration process,

$$\mathbf{N}(\mathbf{X}) = \left\{ \begin{array}{c} \mathbf{u}_1^{\mathrm{T}}\mathbf{u}_1 - 1 \\ \vdots \\ \mathbf{u}_{n-1}^{\mathrm{T}}\mathbf{u}_{n-1} - 1 \end{array} \right\}_{(n-1)} \tag{34}$$

The formulation in Eq. (34) leads to a simpler partial derivative.

Path constraints are included as inequality constraints, which are converted to equality constraints by use of slack variables, a successful numerical adaptation from nonlinear programming.[2,35] The slack variables are incorporated in $\mathbf{X}$ and associated with the other state elements at their particular epoch, i. In this analysis, the elevation angle at each epoch, ϕ_i, and the altitude at each epoch, A_i, are two sample trajectory constraints.[2] Altitude is defined as the distance from the lunar south pole,

$$A_i = \sqrt{(x_i - 1 + \mu)^2 + y_i^2 + (z_i + R_m)^2} \tag{35}$$

The third path constraint requires that the sailface normal, or control $\mathbf{u}_i$, is always directed away from the Sun (the sunlight vector is $\hat{\boldsymbol{\ell}}_i$). Of the inequality constraints, only this attitude requirement is mandated. Together, these three inequality constraints are written as

$$\begin{aligned} \phi_{lb} \leq \phi_i &\equiv \arcsin\left(-\frac{z_i + R_m}{A_i}\right) \\ A_{ub} &\geq A_i \\ 0 &\leq \hat{\boldsymbol{\ell}}_i \cdot \mathbf{u}_i \end{aligned}$$

The associated slack variables are squared and added to the inequality constraints, resulting in the following equality constraints

$$\mathbf{g}_i(\mathbf{r}_i, \boldsymbol{\eta}_i) = \left\{ \begin{array}{l} \sin\phi_{lb} + \frac{z_i + R_m}{A_i} + \eta_{\phi,i}^2 \\ A_i - A_{ub} + \eta_{A,i}^2 \\ -(\hat{\boldsymbol{\ell}}_i \cdot \mathbf{u}_i) + \eta_{\ell,i}^2 \end{array} \right\}_m = \{\mathbf{0}\}_m \tag{36}$$

Combining the path constraints at all epochs results in $m(n-1)$ total elements in $\mathbf{g}(\mathbf{X})$. When using the path constraints in Eq. (36), $m = 3$. Other possible path and attitude constraints are discussed in the Appendix.

Only the first $(n-1)$ epochs are required in formulating $\Delta\mathbf{a}$, $\Delta\mathbf{v}$, $\mathbf{N}$, and $\mathbf{g}(\mathbf{X})$ in the constraint vector $\mathbf{F}(\mathbf{X})$. Due to periodicity, the first and n^{th} epochs are identical. Activating the nth epoch yields a problem that is over-constrained and the Jacobian, $D\mathbf{F}(\mathbf{X})$, is not full rank.

The reader might notice that AFDM algorithms are similar to a constrained least-squares optimization problem. A brief discussion of using the AFDM as part of an optimization problem is presented in the Appendix.

INITIAL GUESSES

Not surprisingly, the initial guess for the trajectory as well as the control history influences the resulting solution. Two types of initial guess are explored for the trajectory and six different types for the initial guess corresponding to the control history. The various combinations are summarized in Table 1. The initial guesses corresponding to the slack variables are always established such that $\mathbf{g}^0(\mathbf{X}) = \mathbf{0}$.*

Due to the required periodicity of the converged orbit, co-axial circles offset from the Moon in the z direction are selected as one option for the orbit to develop the initial guess: the x and y coordinates are defined by simple sinusoidal functions and the z components are constant (cases labeled "C" in Table 1). Associated initial-guess velocities are defined as the time derivatives of these sinusoidal functions. In some cases, the converged trajectories appear similar to their respective offset luni-axial circles; in most trials, however, the converged trajectory does not resemble the initial circle. The other possible option for the development of an initial guess for the trajectory is a simple static point, placed in the xz plane defined for the Earth–Moon CR3B system. This option is denoted as "P" in Table 1. Of course, only the two Lagrange points near the Moon preserve this initial guess as a converged result, and only if the sailface normal is orthogonal to the sunline at all times. Nevertheless, the AFDMs converged on many different solutions using this strategy.

Table 1. Summary of trajectory and control history initial-guess strategies

IG Trajectory	IG Control	Description
C		$\mathbf{r}^0$ is a circle offset from Moon in $-z$ direction
P		$\mathbf{r}^0$ clustered at a point in neighborhood of Moon in southern xz half-plane
	α^*	$\mathbf{u}_i^0$ south from Sun-line by 35.26°
	Moon	$\mathbf{u}_i^0$ away from Moon
	EOM-NR	$\mathbf{u}_i^0$ solves EOM at each epoch, $\|\mathbf{u}_i^0\|$ not necessarily equal to 1
	RAA	$\mathbf{u}_i^0$ is in direction of required applied acceleration at each epoch, $\|\mathbf{u}_i^0\| = 1$
	$\boldsymbol{\nabla} U$	$\mathbf{u}_i^0$ points in the direction of $\boldsymbol{\nabla} U(\mathbf{r}(t_i))$
	$\hat{\boldsymbol{\ell}}$	$\mathbf{u}_i^0$ points along the sunline, $\hat{\boldsymbol{\ell}}(t_i)$

The "control history" refers to the direction of the sailface normal, or applied thrust vector, along the entire orbit. Six concepts are explored as initial guesses for the control history. The basis of the first strategy is a control that maximizes the out-of-plane force contributed by the sail ("α^*" in Table 1). Derived analytically by McInnes [32, (pp. 115–118, 223–224)], the sail-pitch angle that maximizes the out-of-plane thrust is $\alpha^* = 35.26^\circ$. The initial guess for the thrust vector is then

$$\mathbf{u}_i^0 = \cos(\Omega t_i)\cos\alpha^*\hat{\mathbf{b}}_1 - \sin(\Omega t_i)\cos\alpha^*\hat{\mathbf{b}}_2 + \sin\alpha^*\hat{\mathbf{b}}_3 \tag{37}$$

This initial guess for the control strategy, combined with a point initial guess for the trajectory is very successfully applied by Ozimek et al.[2]

*The superscript "0" will indicate an initial guess, "f" will indicate a converged solution.

The next strategy that serves as an initial guess option for the control is one where the thrust vector is directed away from the center of the Moon throughout the initial guess trajectory ("Moon" in Table 1). Thus, if a vector, $\mathbf{r}_i^c$, is defined as the difference between the position of the Moon and the position of the spacecraft in the rotating frame at time i, or

$$\mathbf{r}_i^c = (x_i - 1 + \mu)\hat{\mathbf{b}}_1 + y_i\hat{\mathbf{b}}_2 + z_i\hat{\mathbf{b}}_3 \tag{38}$$

then the initial guess for the control strategy is

$$\mathbf{u}_i^0 = \frac{\mathbf{r}_i^c}{\|\mathbf{r}_i^c\|} \tag{39}$$

By definition, $\mathbf{u}_i^0$ possesses unit magnitude.

A third initial control strategy is one that solves the equations of motion, Eq. (1), at each epoch for $\mathbf{u}_i^0$ with the assumption that the initial guess trajectory is correct, that is,

$$f(\mathbf{u}_i^0) = \widetilde{\mathbf{a}}_i + 2\left(\boldsymbol{\omega} \times \widetilde{\mathbf{v}}_i\right) + \boldsymbol{\nabla} U(\mathbf{r}_i) - \beta(\hat{\boldsymbol{\ell}}(t_i) \cdot \mathbf{u}_i^0)^2 \mathbf{u}_i^0 = 0 \tag{40}$$

A Newton-Raphson iteration scheme is used to solve this nonlinear equation, with each $\mathbf{u}_i^0$ initially directed away from the Moon. The converged control history then serves as the input control history, $\mathbf{u}^0$, to the AFDMs. Unlike the alternative strategies, the control vector is not unitized, and the AFDMs are expected to render a viable solution. This strategy is labeled "EOM-NR" in Table 1.

A fourth, simpler initial control strategy assumes that the initial guess trajectory is already a solution to the equations of motion, Eq. (1), with the caveat that the sail provides any additional, required applied acceleration without regard to feasibility or practicality (i.e., the sail is assumed capable of unlimited, variable thrust, independent of direction). Therefore, if the applied acceleration that is required to solve the equations of motion appears as

$$\mathbf{a}_{rapp,i} = \widetilde{\mathbf{a}}_i + 2\left({}^e\boldsymbol{\omega}^b \times \widetilde{\mathbf{v}}_i\right) + \boldsymbol{\nabla} U(\mathbf{r}_i) \tag{41}$$

then the initial guess for the control strategy is

$$\mathbf{u}_i^0 = \frac{\mathbf{a}_{rapp,i}}{\|\mathbf{a}_{rapp,i}\|} \tag{42}$$

This strategy is labeled "RAA" for "required applied acceleration" in Table 1.

The fifth option for the initial guess for the control history (labeled "$\boldsymbol{\nabla} U$" in Table 1) is similar to the concept presented in Eqs. (41) and (42), except that only the pseudo-gravity gradient, $\boldsymbol{\nabla} U(\mathbf{r})$, is involved, that is,

$$\mathbf{u}_i^0 = \frac{\boldsymbol{\nabla} U(\mathbf{r})}{\|\boldsymbol{\nabla} U(\mathbf{r})\|} \tag{43}$$

Of course, if the sailcraft moves, as is the case where the initial guess for the trajectory is a circle, the relative and Coriolis acceleration terms are non-zero. The pseudo-gravity gradient for the control direction is a critical quantity to develop equilibrium surfaces in the problem and may yield some insight (e.g., McInnes [32, pp. 196–224]).

The remaining initial-guess concept for the control history is completely independent of the initial guess associated with the trajectory. The initial control strategy assumes that the sailface normal is parallel to the sunlight direction (labeled "$\hat{\boldsymbol{\ell}}$" in Table 1).

Through the iterative Newton-Raphson process, the directions of the sailface normal at the nodes along the trajectory, as well as the path itself, align to solve the equations of motion. It should be noted, however, that the number of nodes, n, affects whether or not *any* solutions will converge when the initial guess for the path was a point combined with an initial guess for the control history other than the α^* or $\hat{\boldsymbol{\ell}}$ strategies. This lack of convergence for these initial guess combinations occurs when $n - 1$ is a multiple of 4, indicating that the geometry of the node placement plays a significant role in convergence.

RESULTS

The primary purpose of using a simple, lower-fidelity, augmented finite difference method is to quickly and easily examine the design space for a solar-sail spacecraft in orbit near the Moon. Different combinations of initial guess lead to different solutions. This is especially true when when comparing orbits that arise from circular initial guesses for the path to those that arise from initial guesses that collapse the trajectory to a single point. A general survey of the solution space for all combinations of initial guesses is a first step in understanding the design space. A viable orbit from any method possesses characteristics that can be used to categorize the orbit. Orbits from an AFDM method should reasonably approximate orbits from a higher-accuracy method. To validate the results of the AFDM algorithms, the solution from an AFDM process are used as inputs to a Hermite-Simpson collocation scheme. Of course, the orbits from the higher-accuracy scheme also reflect actual, viable trajectories. These topics will be examined in the following subsections.

Survey of Solutions: Examining the Design Space

Once the AFDM algorithms are developed, they are used in an extensive survey of the solar sail design space in the CR3B Earth–Moon system. A large variety of sail characteristic accelerations are tested in combination with different initial guesses for both the trajectory, the control history (described in the previous section), and path constraints.

The highest published value for the characteristic acceleration, a_0, associated with a designed sail is 2.26 mm/s^2,[36] but the highest value consistent with a recent technology demonstration is closer to 0.58 mm/s^2.[33] Although characteristic accelerations above these values are unrealistic, higher values of a_0 are incorporated to demonstrate the process. In cases where the initial guess for the path is a point, a grid spanning -75000 to 75000 km in the x direction and -75000 to 0 km in the z direction, each in 1000 km increments are used for x^0 and z^0 for all points along the path (11476 runs per grid). When the initial guess for the path is a concentric circle below the Moon, a grid of radii and z-offset values span a region from 0 to 75000 km and -75000 to 0 km, respectively, at 1000 km increments (5776 runs per grid). The origin of the coordinate system is the center of the Moon for both sets of runs. Finally, different sets of path constraints are also examined. For the lunar communications relay, an elevation constraint of $\phi_{lb} = 15^\circ$ and altitude constraint of $A_{ub} = 384400$ km is assumed. For these constraints, the sail's characteristic acceleration must be at least $a_0 = 1.5$ mm/s^2 to yield convergence, as smaller values of a_0 are insufficient to push the vehicle far enough south of the Moon to satisfy the elevation constraint. However, when the elevation constraint is relaxed to 6° or 4°, solutions emerge for a sail with a smaller acceleration requirement, $a_0 = 0.58$ mm/s^2; such trajectories exist below L_2 and L_1, respectively.

Varying the initial conditions, the characteristic accelerations, and some of the constraints leads to millions of sample orbits for both AFDMs. This massive number of simulations merits parallelization. Multiple dual quad-core machines are employed with the Matlab Distributed Computing Toolbox (DCT).[34] The DCT is limited to four cores per Matlab session, so two Matlab sessions were invoked on each machine. The main function nested initial conditions (x^0, or r^0, and z^0) in loops of a_0; the constraint values, initial-guess trajectory strategy, and initial-guess control strategy were inputs to the function. On average, and depending on how many solutions actually converged, an AFDM-RV run using the point initial-guess strategy required 20 to 24 hours and a run using the luni-axial circle initial-strategy is completed in 9 to 12 hours (twice as many initial conditions were evaluated in the point cases than in the circle cases). The Jacobian, $D\mathbf{F}(\mathbf{X})$, in the AFDM-R is 75% of the size of the Jacobian in the AFDM-R, and the runs using AFDM-R took about 75% of the time for the AFDM-RV. Frequently called functions were rewritten as C-code and compiled as MEX-files for use in Matlab.[34] Further speed improvements could result from rewriting and compiling larger blocks of code.

In general, an initial guess for the control history that relies on the α^* strategy is most likely to converge. The $\hat{\boldsymbol{\ell}}$ strategy, where the sailface normal is initially directed away from the Sun throughout the trajectory, is the next most likely strategy to lead to convergence. The remaining four control strategies in Table 1 were more likely to converge when compared to the α^* and $\hat{\boldsymbol{\ell}}$ strategies for high characteristic accelerations (i.e., $a_0 > 5$ mm/s^2). It should not be assumed that one initial control strategy is necessarily superior to any other,

even if that strategy converges more often. An appropriate plan for examining the design space is to select a characteristic acceleration and path constraints, and then generate solutions based on multiple combinations of initial guesses for the path and control for one or both of the two AFDM algorithms. This is especially true for large characteristic accelerations.

Due to memory limitations, only a partial set of data, consisting of the first state along the trajectory and other significant metrics, is stored. The first state is defined to occur when the Sun is located along the negative x-axis. Each trajectory is represented by its respective first point along the initial guess trajectory and the corresponding first point along a converged trajectory. The collection of first points corresponding to either the initial-guess or converged trajectory form a section. A sample pair of these sections appear in Fig. 2. The initial guess strategy for the trajectories represented in these plots assumes offset luni-axial circles for the path and a sail orientation angle α^* from Eq. (37); constraints are equal to $\phi_{lb} = 15^\circ$ and $A_{ub} = 384400$ km. Each colored dot represents the first point along the particular trajectory, and the color links the initial guesses to the solutions. Where there is no color on the left plot, that initial guess combination did not converge (e.g., a 50,000 km radius circle offset 10,000 km from the Moon). Four example orbits illustrate this convergence process. Note that the converged blue orbit in Fig. 2 is much larger than and far from its initial guess.

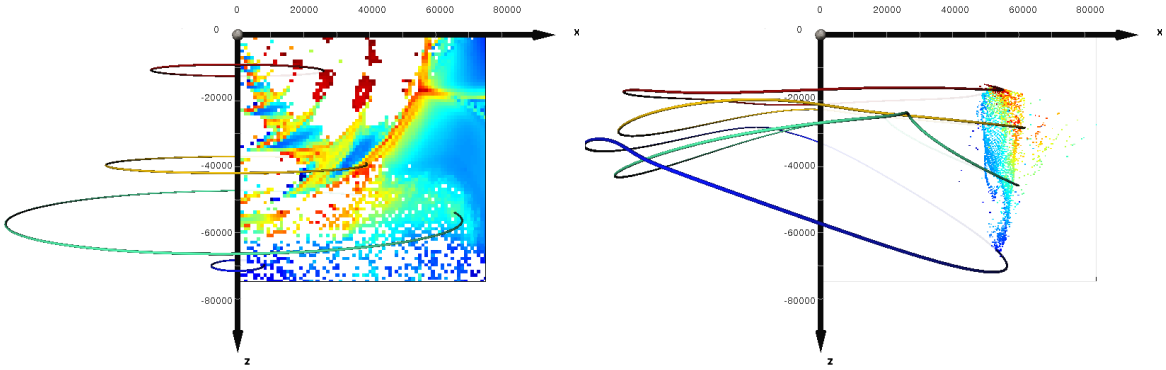
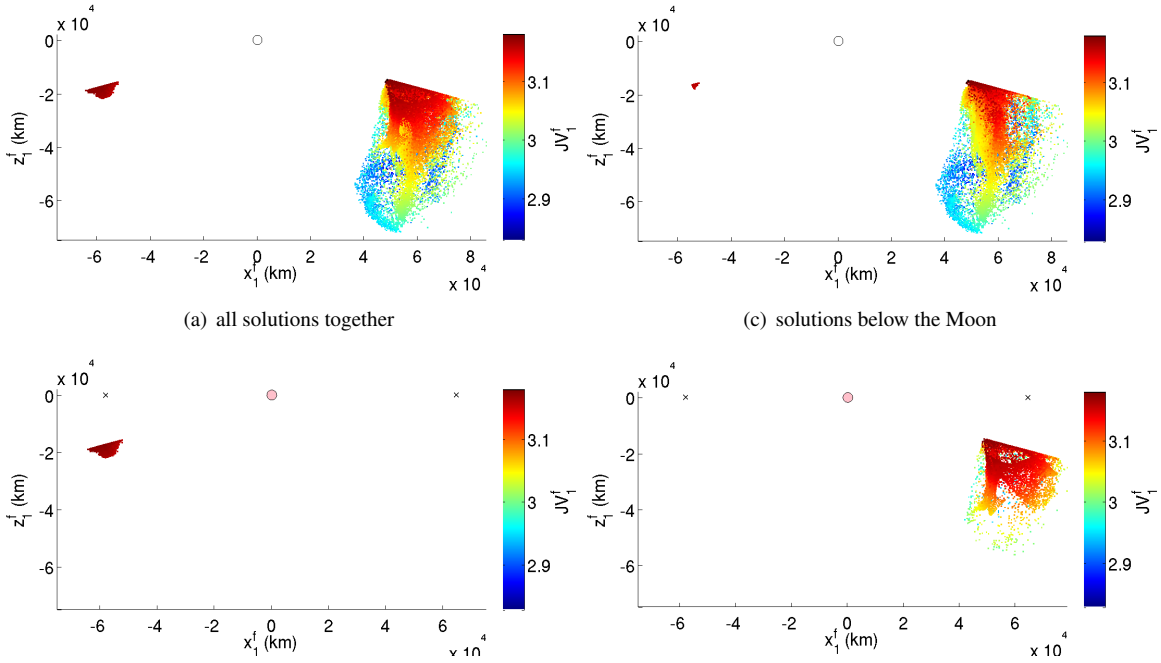


Figure 2 Sections of initial guesses (left) and converged solutions (right) for one initial guess strategy. The Moon is at the origin.

For a set of initial guess strategies pertaining to a particular characteristic acceleration, all results can be combined and sorted into orbit that exists below L_1, the Moon, or L_2. The first-point sections for all solutions in the 1.70 mm/s^2 case are plotted in Fig. 3(a), and Fig. 3(b–d) shows the first-point sections separated by these different regions. The 15° elevation constraint is apparent in these plots. To differentiate the first points from each other, the expression for the Jacobi constant is adapted and evaluated at each point, providing a color scale. Of course, this value is not constant throughout the trajectory, but does provide qualitative information on the energy of the orbit over one period. The dots in Fig. 3 do not comprise all of the possible solutions in the design space. They are only the ones that converge using the initial condition strategies. Solutions from other, untested initial condition strategies also likely exist. However, by testing the previously defined twelve combinations of initial guess options (path and control) and two AFDM algorithms, a broad picture of the solution space emerges.

Recall that the Sun moves in a retrograde fashion with respect to the fixed Earth–Moon system. In general, the solutions that orbit below the Moon also move retrograde and start in opposition to the Sun (i.e., on the right-hand side of the xz-plane; the few seen on the left-hand side in Fig. 3-(c) are prograde orbits, starting in conjunction). Solutions that orbit below either Lagrange point can be either prograde or retrograde. All prograde orbits under L_1 and most under L_2 have their first point along the orbits in conjunction with the Sun with respect to the center of the orbit (i.e., further left in the xz-plane). Retrograde orbits in the vicinity of the Lagrange points do exist in this survey, and they all have their first point along the orbits in opposition to the Sun with respect to the center of the orbit (i.e., further right in the xz-plane). Retrograde motion in



(a) all solutions together

(c) solutions below the Moon

(b) solutions below L_1

(d) solutions below L_2

Figure 3 First-point sections of all solutions in the 1.70 mm/s^2 case by segregated by location. Path constraints are $\phi_{lb} = 15°$ and $A_{ub} = 384400$ km.

the vicinity of the Lagrange points is consistent with motion ascertained through linear analysis resulting in planar Liapunov orbits in the circular restricted three-body problem. Spacecraft in retrograde orbits under either Lagrange point or the Moon are always on the opposite side the orbit from the Sun.

Dynamical Characteristics of Offset Lunar Orbits

The solutions can be further characterized by the profile of the constant from respective Jacobi integrals throughout the orbit as an attempt to group orbits by their common dynamical structures. As with the differentiation of the first points in the section plots (Figs. 2 and 3), the expression for the Jacobi constant is adapted to aid in characterization. Because the sail provides a thrust altering the energy of the system, this energy-like quantity is not constant for any length of time. This categorization, based on a fourth-degree polynomial fit to the profile of the Jacobi "constant" along the orbit, is rudimentary, but does provide some insight into the different types of trajectories.

The vast majority orbits centered on the Moon follow a retrograde path, and the values of the Jacobi integrals throughout the orbits can be categorized into four profiles. The first appears as a "W" when plotted and is characterized by relatively high Jacobi values when the spacecraft is near the x-axis, which reflects a decreased rotating velocity. Typically, these orbits are close being planar, circular, offset orbits that are parallel to the xy plane. Their shape sometimes includes a "pinch" or "swirl" in the orbit near the x-axis when the characteristic acceleration of the sail is above 2 mm/s^2. For these larger characteristic accelerations, the spacecraft orbits closer to the Moon and the swirl is a way to allow the period of the orbit to equal that of a synodic month. In other cases involving a W-profile, the orbits appear "bend" up along the y-axis. Subfigures (a–d) in Fig. 4 highlight three example orbits in the Earth–Moon rotating frame from the Moon-centered, W-profile group. The arrows in the orbit plots represent the direction of the sailface normal (i.e., the control) at that state along the trajectory. The states are at evenly spaced intervals, therefore, arrows that are closer together indicate that the spacecraft moves more slowly along the orbit during those intervals. The slower motion corresponds to higher Jacobi values. The second Jacobi profile is that of an inverted "V," which will be

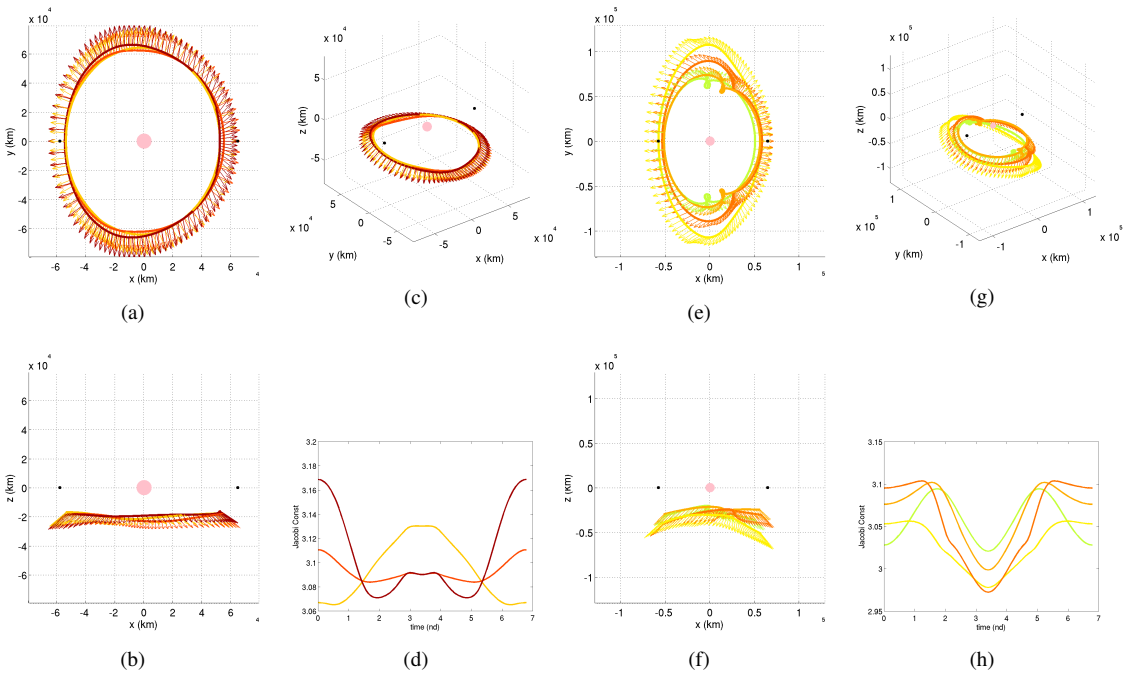


Figure 4 Examples from the Moon-centered, W-profile (a–d) and M-profile (e–f) groups. The black dots are the L_1 and L_2 points, and the Moon is enlarged by a factor of three. Arrows along the trajectories mark the direction of the sail normal at that point. The "W" and "M" shapes are apparent in the Jacobi profiles, subfigures (d) and (h), respectively.

labeled "A." Its characteristics are similar to that of the "W" profile orbits and could be considered a sub-class of the "W" orbits.

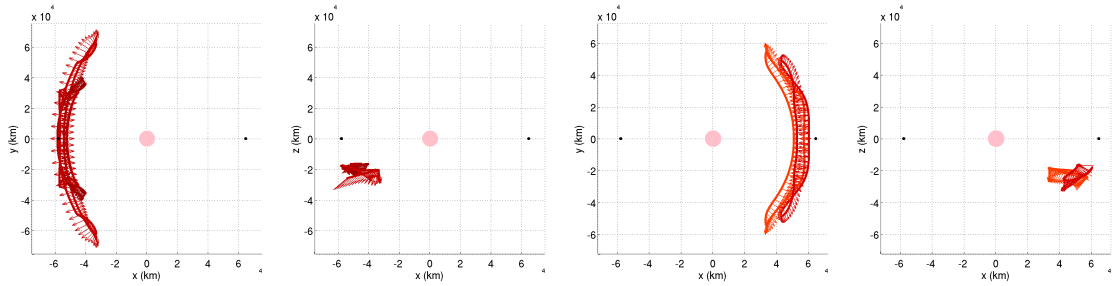
The other dominant profile for orbits below the Moon appears "M" shaped. The orbits are typically characterized by a wide shape and folded down along the y-axis or rounder orbits with twist along the y-axis. Like the "W" orbits, the twist allows the orbit to match the lunar synodic period. Five examples are shown in Fig. 4 in subfigures (e–h). The final profile is "V" shaped and is similar to the "M" orbits.

Likewise, orbits under L_1 and L_2 exhibit similar profiles based on the polynomial fit, with "W" being the dominant profile. However, since the locations of these orbits under the two Lagrange points and Moon differ, a comparison of W, A, M, or V orbits between differently centered orbits is not appropriate. Unlike the Moon-centered orbits, the converged solutions from the survey of the solutions space that exist under the Lagrange points are more often prograde then retrograde. Examples of each are shown in Figs. 5 and 6. Like the orbits in the other figures, the color reflects the value of the Jacobi integral at the first point along the trajectory. Higher values are observed in the L_1 retrograde and L_2 prograde orbits and reflect the velocity required to maintain these orbits in the rotating frame.

It should also be noted that the maximum turn rates for the orbits presented in these figures are all approximately 1 deg/hr or less and the maximum yaw and pitch accelerations are all less than 1×10^{-8} deg/s^2, which is consistent with recent analyses for a solar sail demonstration mission.[37]

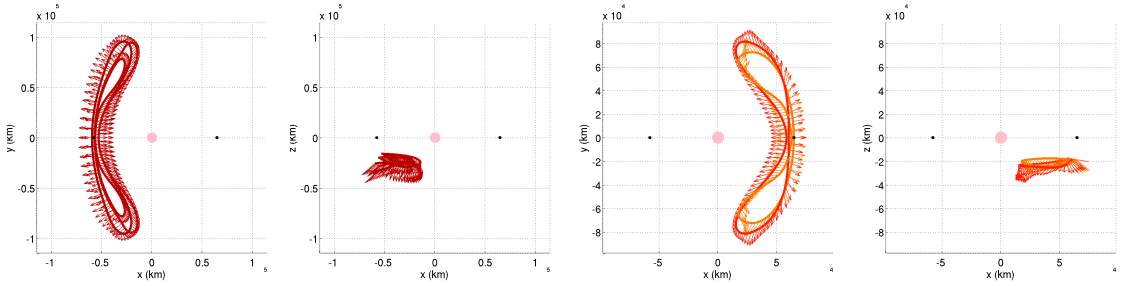
AFDMs Compared to Hermite-Simpson Collocation

As stated, prior to any analysis, the design space for this problem was unknown and after a survey based on the AFDM algorithms, a set of solutions accurate to $\mathcal{O}(\Delta t^2)$ emerges. These solutions can be further employed to refine the design space by selecting them as initial guesses for a higher-accuracy method, such



(a) (b) (c) (d)

Figure 5 Prograde, offset, Lagrange-point orbits under L_1 (a,b) and L_2 (c,d); xy-(a,c) and xz-views (b,d) are shown for each. The Moon is three times scale, and the black dots represent the locations of the Lagrange points.



(a) (b) (c) (d)

Figure 6 Retrograde, offset, Lagrange-point orbits under L_1 (a,b) and L_2 (c,d); xy-(a,c) and xz-views (b,d) are shown for each. The Moon is three times scale, and the black dots represent the locations of the Lagrange points.

as Hermite-Simpson collocation. Based on work by Ozimek et al.,[2] a crude Hermite-Simpson collocation method was developed to test the AFDM solutions. There is one noteworthy difference between the Hermite-Simpson collocation method employed by Ozimek et al. and the adaptation used for this analysis. Ozimek et al. employ a small step along the imaginary axis to calculate the partial derivatives related to the defects. For the current analysis, a simple central difference approximation (CDA) represents those partial derivatives. Both are numerical schemes for computing difficult partial derivatives and subject to roundoff errors, but the CDA is also subject to truncation error, while the imaginary-step method is not. Thus, the error for the CDA is minimized at a step size greater than zero (on the order of 10^{-8}), but the error for the imaginary-step method approaches the roundoff error as the step size goes to zero (on the order of machine precision).[38]

This re-convergence test is conducted on a subset of solutions from the AFDM analysis to demonstrate the suitability of those solutions as inputs for the collocation scheme. Constraints are fixed at $\phi_{lb} = 15^\circ$ and $A_{ub} = 384400$ km for both schemes. Better convergence is achieved with the collocation procedure where the characteristic acceleration is lower. A re-convergence rate of approximately 80% is observed for $a_0 = 1.7$ mm/s^2 under the specified constraints. The lower re-convergence rates for higher characteristic accelerations might reflect the more restrictive dynamics involved with those higher accelerations. The AFDM may converge on a solution whose finite-difference and evaluated accelerations match to second order, but the AFDM provide insufficient information about higher-order consistency. A converged solution from the AFDM may not reflect an actual solution in some cases. The limited accuracy of the partial derivatives incorporated in the Hermite-Simpson collocation scheme could also mask the subtleties of the convergence process. Further investigation is warranted. The maximum differences between the converged AFDM solutions and re-converged Hermite-Simpson solutions at a selected node along each trajectory appears in Fig. 7

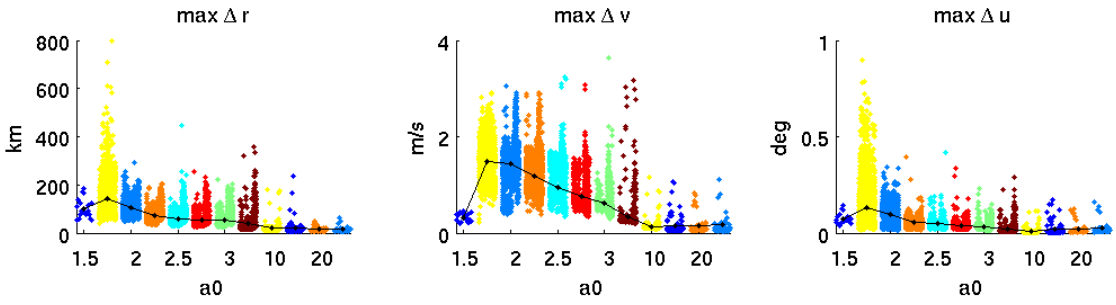


Figure 7 Maximum deviations in position, velocity and control for a variety of characteristic accelerations (a_0 in units of mm/s^2). The black line represents the average maximum deviation within a set of trajectories for that characteristic acceleration.

for a number of characteristic accelerations. For example, for all re-converged trajectories that are defined for $a_0 = 1.7$ mm/s^2 (in yellow), the largest difference in position at a node along the trajectory is approximately 800 km, well within the 1700 km range derived in the error analysis. Maximum deviations in velocity and control are also indicated.

CONCLUDING REMARKS

It is possible to achieve an offset orbit in view of the lunar south pole at all times using a solar sail. However, to meet visibility requirements from the south pole, the sail must have a characteristic acceleration sufficiently large to provide the necessary thrust.

Designing an appropriate trajectory in the dynamically complicated Earth–Moon–Sun regime is not straightforward. Fortunately, numerical methods to solve boundary value problems can be employed to generate viable trajectories. Augmented finite-difference methods are examples of these numerical tools and have the advantage of being simple to develop and understand. Solutions resulting from these algorithms can further serve as initial guess options for more accurate numerical methods. Using the AFDM algorithms, a broad survey of initial conditions exposes a large number of viable trajectories to the mission designer. The trajectories from this survey possess dynamical characteristics that can serve as an initial attempt to classify the trajectories. Further work on understanding the underlying dynamical structure is warranted.

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APPENDIX

Using the AFDMs in a Trajectory Optimization Problem

The AFDM algorithms are similar to a constrained least-squares optimization problem, where the objective (based on the AFDM-RV algorithm) is

$$\min \|f(\mathbf{X})\| = \left(\sum_{i=1}^{n-1} \left((\mathbf{a}_i - \widetilde{\mathbf{a}}_i)^2 + (\mathbf{v}_i - \widetilde{\mathbf{v}}_i)^2 \right) \right)^{1/2} \tag{44}$$

subject to

$$\begin{aligned} \phi_{lb} - \arcsin\left(-\frac{z_i + R_m}{A_i}\right) \leq 0, \qquad & i = 1, \ldots, n-1 \\ A_i - A_{ub} \leq 0, \qquad & i = 1, \ldots, n-1 \\ -\hat{\boldsymbol{\ell}}_i \cdot \mathbf{u}_i \leq 0, \qquad & i = 1, \ldots, n-1 \end{aligned} \tag{45}$$

and equality constraints

$$\begin{aligned} \mathbf{r}_1 &= \mathbf{r}_n \\ \mathbf{v}_1 &= \mathbf{v}_n \\ \mathbf{u}_1 &= \mathbf{u}_n \\ \mathbf{u}_i^{\mathrm{T}}\mathbf{u}_i - 1 = 0, \qquad & i = 1, \ldots, n-1 \end{aligned} \tag{46}$$

The $(\mathbf{v}_i - \widetilde{\mathbf{v}}_i)$ term in Eq. (44) is dropped if the optimization problem is based on the AFDM-R algorithm. Either formulation is amenable to SQP and uses no slack variables.

Alternatively, a mission designer may want to minimize the characteristic acceleration of the solar sail. In that case, the objective function becomes

$$\min J = a_0 \tag{47}$$

subject to the constraints in Eqs. (45) and (46), with the additional equality constraints of

$$\mathbf{a}_i - \widetilde{\mathbf{a}}_i = 0, \qquad i = 1, \ldots, n-1 \tag{48}$$

$$\mathbf{v}_i - \widetilde{\mathbf{v}}_i = 0, \qquad i = 1, \ldots, n-1 \tag{49}$$

Equation (49) is only present if the formulation to solve the problem is based on the AFDM-RV algorithm.

Additional Path Constraints

Two path constraints and one attitude constraint were discussed previously. Other constraints that could be useful for mission design and mission planning would include a lower bound on altitude, a maximum spacecraft turn rate, and a ground antenna slew rate. The lower bound on altitude would be similar in construction to the upper bound on altitude, except that the slack variable would be a surplus variable. A spacecraft turn-rate constraint is useful because solar sails are expected to reorient at relatively slow turn rates.

The turn angle of a solar sail can be expressed using the spherical law of cosines, that is,

$$\Delta\sigma = \arccos(\cos\phi_i \cos\phi_{i+1} \cos\Delta\lambda + \sin\phi_i \sin\phi_{i+1}) \tag{50}$$

where $\phi_{i,i+1}$ and $\lambda_{i,i+1}$ is direction of the sailface normal before and after the turn using spherical coordinates latitude (ϕ) and longitude (λ). The turn-rate constraint is then $\dot{\sigma} \leq \dot{\sigma}_{max}$, or, if $\dot{\sigma} = \Delta\sigma/\Delta t$ and making use of a slack variables at each epoch

$$\Delta\sigma_i - \dot{\sigma}_{max}\Delta t + \eta_{\sigma,i} = 0 \tag{51}$$

Eq.(50) can be written in terms of the unit-length pointing vector, $\{\xi, \gamma, \zeta\}^{\mathrm{T}}$, in the inertial frame, that is,

$$\Delta\sigma = \arccos(\xi_i\xi_{i+1} + \gamma_i\gamma_{i+1} + \zeta_i\zeta_{i+1}) \tag{52}$$

and the partial derivatives are thus

$$\begin{aligned} \frac{\partial\Delta\sigma}{\partial\xi_i} &= -\frac{\xi_{i+1}}{\sqrt{1-(\xi_i\xi_{i+1}+\gamma_i\gamma_{i+1}+\zeta_i\zeta_{i+1})^2}}, & \frac{\partial\Delta\sigma}{\partial\xi_{i+1}} &= -\frac{\xi_i}{\sqrt{1-(\xi_i\xi_{i+1}+\gamma_i\gamma_{i+1}+\zeta_i\zeta_{i+1})^2}} \\ \frac{\partial\Delta\sigma}{\partial\gamma_i} &= -\frac{\gamma_{i+1}}{\sqrt{1-(\xi_i\xi_{i+1}+\gamma_i\gamma_{i+1}+\zeta_i\zeta_{i+1})^2}}, & \frac{\partial\Delta\sigma}{\partial\gamma_{i+1}} &= -\frac{\gamma_i}{\sqrt{1-(\xi_i\xi_{i+1}+\gamma_i\gamma_{i+1}+\zeta_i\zeta_{i+1})^2}} \\ \frac{\partial\Delta\sigma}{\partial\zeta_i} &= -\frac{\zeta_{i+1}}{\sqrt{1-(\xi_i\xi_{i+1}+\gamma_i\gamma_{i+1}+\zeta_i\zeta_{i+1})^2}}, & \frac{\partial\Delta\sigma}{\partial\zeta_{i+1}} &= -\frac{\zeta_i}{\sqrt{1-(\xi_i\xi_{i+1}+\gamma_i\gamma_{i+1}+\zeta_i\zeta_{i+1})^2}} \end{aligned} \tag{53}$$

Note that the denominators in Eq. (53) approach zero when $\Delta\sigma \ll 1$, or when the pointing vector does not change direction in an inertial frame between epochs. Fortunately for this problem, the direction of the sail normal is always changing to compensate for the motion of the primary bodies.

Ground antenna slew-rate constraints can also be expressed using spherical coordinates. The analysis for incorporating those constraints is similar to the spacecraft turn-rate constraint.

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